NOTES ON SOLUTION TO THE BUOYANCY EQUATION WITH EXPONENTIAL κ PROFILE

GPT 5.5 WITH PROMPTING FROM TWNH

1. Problem statement

Consider the one-dimensional vertical boundary-value problem

$$Ek\,\partial_z(\kappa(z)\,\partial_z b) = \Gamma\,(b - B)\,, \tag{1}$$

on the interval

$$-H \le z \le 0,$$

with boundary conditions

$$b(0) = b_s, \tag{2}$$

$$\partial_z b(-H) = 0. \tag{3}$$

Here $B = B(x, y, z)$ is a prescribed buoyancy profile, $b_s = b_s(x, y)$ is the imposed surface buoyancy, Γ is a relaxation coefficient, and Ek is a constant prefactor.

We take the vertical diffusivity to be

$$\kappa(z) = a + \beta \exp\left[-\frac{z + H}{h}\right], \tag{4}$$

where $a > 0$, $h > 0$, and β is a constant. I use the symbol β for the exponential coefficient in κ to avoid notational conflict with the buoyancy solution b.

The horizontal dependence x, y is parametric in this vertical problem, so it will be suppressed in most of the derivation.

Define

$$\mu = \frac{\Gamma}{Ek}. \tag{5}$$

Then (70) may be written as

$$\partial_z(\kappa b_z) - \mu b = -\mu B. \tag{6}$$

This is a forced Sturm–Liouville problem with mixed Dirichlet–Neumann boundary conditions.

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2. Homogeneous equation

The associated homogeneous equation is

$$\partial_z(\kappa b_z) - \mu b = 0. \tag{7}$$

Using

$$\kappa(z) = a + \beta \exp\left[-\frac{z+H}{h}\right],$$

introduce the transformed variable

$$X(z) = -\frac{\beta}{a}\exp\left[-\frac{z+H}{h}\right]. \tag{8}$$

Then

$$\kappa(z) = a(1-X). \tag{9}$$

Also define the dimensionless parameter

$$\nu = h\sqrt{\frac{\mu}{a}} = h\sqrt{\frac{\Gamma}{Ek\,a}}. \tag{10}$$

With these definitions, a convenient basis of homogeneous solutions is

$$y_+(z) = X(z)^{\nu}\,{}_2F_1(\nu, \nu+1; 2\nu+1; X(z)), \tag{11}$$

$$y_-(z) = X(z)^{-\nu}\,{}_2F_1(-\nu, 1-\nu; 1-2\nu; X(z)). \tag{12}$$

Therefore the general homogeneous solution is

$$b_h(z) = C_+ y_+(z) + C_- y_-(z). \tag{13}$$

The function ${}_2F_1$ is the Gauss hypergeometric function. When $\beta > 0$, one has $X < 0$, and the real-valued branch should be chosen consistently.

3. Construction of boundary-adapted homogeneous solutions

We now construct two special homogeneous solutions.

Let $\phi(z)$ solve

$$L\phi = 0, \qquad \phi_z(-H) = 0, \tag{14}$$

where

$$Lf := \partial_z(\kappa f_z) - \mu f. \tag{15}$$

Similarly, let $\psi(z)$ solve

$$L\psi = 0, \qquad \psi(0) = 0. \tag{16}$$

Using the homogeneous basis $y_\pm$, a convenient choice is

$$\phi(z) = y_-'(-H)y_+(z) - y_+'(-H)y_-(z), \tag{17}$$

because

$$\phi_z(-H) = y_-'(-H)y_+'(-H) - y_+'(-H)y_-'(-H) = 0.$$

Likewise, a convenient choice for ψ is

$$\psi(z) = y_-(0)y_+(z) - y_+(0)y_-(z), \tag{18}$$

because

$$\psi(0) = y_-(0)y_+(0) - y_+(0)y_-(0) = 0.$$

4. Green's function

The Green's function $G(z,\zeta)$ is defined by

$$L_z G(z,\zeta) = \delta(z-\zeta), \tag{19}$$

with boundary conditions

$$G(0,\zeta) = 0, \tag{20}$$

$$\partial_z G(-H,\zeta) = 0. \tag{21}$$

For a Sturm–Liouville operator of the form

$$Lf = \partial_z(\kappa f_z) - \mu f,$$

the weighted Wronskian

$$W = \kappa(z)\left[\phi(z)\psi_z(z) - \phi_z(z)\psi(z)\right] \tag{22}$$

is independent of z.

The Green's function is then

$$G(z,\zeta) = \begin{cases} \dfrac{\phi(z)\psi(\zeta)}{W}, & z < \zeta, \\ \dfrac{\phi(\zeta)\psi(z)}{W}, & z > \zeta. \end{cases} \tag{23}$$

This convention corresponds to the jump condition

$$\kappa(\zeta)\left[G_z(\zeta^+,\zeta) - G_z(\zeta^-,\zeta)\right] = 1. \tag{24}$$

5. Solution of the forced problem

The inhomogeneous problem is

$$Lb = -\mu B, \tag{25}$$

with

$$b(0) = b_s, \qquad b_z(-H) = 0.$$

The boundary condition at the surface is nonhomogeneous, so we decompose the solution into a boundary-forced part and an internally forced part.

The solution satisfying the surface condition $b(0) = b_s$ and the bottom Neumann condition is

$$b_{\mathrm{surf}}(z) = b_s \frac{\phi(z)}{\phi(0)}. \tag{26}$$

The solution forced by the interior source $-\mu B$ and satisfying homogeneous boundary conditions is

$$b_{\mathrm{int}}(z) = -\mu \int_{-H}^{0} G(z,\zeta)B(\zeta)\,d\zeta. \tag{27}$$

Therefore the full solution is

$$\boxed{b(z) = b_s \frac{\phi(z)}{\phi(0)} - \mu \int_{-H}^{0} G(z,\zeta)B(\zeta)\,d\zeta} \tag{28}$$

or, restoring the horizontal dependence,

$$\boxed{b(x,y,z) = b_s(x,y)\,\frac{\phi(z)}{\phi(0)} - \frac{\Gamma}{Ek}\int_{-H}^{0} G(z,\zeta)B(x,y,\zeta)\,d\zeta.} \tag{29}$$

6. Summary of definitions

The diffusivity is

$$\kappa(z) = a + \beta \exp\left[-\frac{z+H}{h}\right]. \tag{30}$$

Define

$$\mu = \frac{\Gamma}{Ek}, \qquad \nu = h\sqrt{\frac{\mu}{a}} = h\sqrt{\frac{\Gamma}{Ek\,a}}, \tag{31}$$

and

$$X(z) = -\frac{\beta}{a}\exp\left[-\frac{z+H}{h}\right]. \tag{32}$$

The homogeneous basis functions are

$$y_+(z) = X(z)^{\nu}\,{}_2F_1(\nu,\nu+1;2\nu+1;X(z))\,, \tag{33}$$

$$y_-(z) = X(z)^{-\nu}\,{}_2F_1(-\nu,1-\nu;1-2\nu;X(z))\,. \tag{34}$$

The bottom-adapted and surface-adapted homogeneous solutions are

$$\phi(z) = y_-'(-H)y_+(z) - y_+'(-H)y_-(z), \tag{35}$$

$$\psi(z) = y_-(0)y_+(z) - y_+(0)y_-(z). \tag{36}$$

The weighted Wronskian is

$$W = \kappa(z)\left[\phi(z)\psi_z(z) - \phi_z(z)\psi(z)\right], \tag{37}$$

which is independent of z.

The Green's function is

$$G(z,\zeta) = \begin{cases} \dfrac{\phi(z)\psi(\zeta)}{W}, & z < \zeta, \\[2ex] \dfrac{\phi(\zeta)\psi(z)}{W}, & z > \zeta. \end{cases} \tag{38}$$

Thus,

$$\boxed{b(x,y,z) = b_s(x,y)\frac{\phi(z)}{\phi(0)} - \frac{\Gamma}{Ek}\int_{-H}^{0} G(z,\zeta)B(x,y,\zeta)\,d\zeta.} \tag{39}$$

7. Constant-diffusivity limit

If $\beta = 0$, then $\kappa = a$, and the governing equation becomes

$$ab_{zz} - \mu b = -\mu B. \tag{40}$$

Define

$$\lambda = \sqrt{\frac{\mu}{a}} = \sqrt{\frac{\Gamma}{Ek\,a}}. \tag{41}$$

The homogeneous solutions are simply

$$b_h(z) = C_1 \cosh[\lambda(z+H)] + C_2 \sinh[\lambda(z+H)]\,. \tag{42}$$

The bottom condition $b_z(-H) = 0$ selects

$$\phi(z) = \cosh[\lambda(z+H)]\,, \tag{43}$$

while the surface-vanishing solution may be chosen as

$$\psi(z) = \sinh(\lambda z), \tag{44}$$

up to an arbitrary multiplicative constant. The same Green's-function construction applies.

This limiting case is useful for checking signs and normalization.

8. Remarks

- The operator
$$L = \partial_z(\kappa\partial_z) - \mu$$
is self-adjoint on $[-H, 0]$ with the mixed Dirichlet–Neumann boundary conditions used above.
- The Green's function is symmetric:
$$G(z,\zeta) = G(\zeta, z),$$
provided the same normalization convention is used.
- The horizontal variables x, y enter only through the data $B(x, y, z)$ and $b_s(x, y)$. For each fixed (x, y), the vertical problem is independent.
- The formula assumes $a > 0$ and $\kappa(z) > 0$ on $[-H, 0]$. Thus one should require
$$a + \beta \exp\left[-\frac{z+H}{h}\right] > 0$$
throughout the water column.

References

[1] NIST Digital Library of Mathematical Functions. *Gauss Hypergeometric Function*. Sections 15.2–15.10. `https://dlmf.nist.gov/15`

[2] F. W. J. Olver, D. W. Lozier, R. F. Boisvert, and C. W. Clark, editors. *NIST Handbook of Mathematical Functions*. Cambridge University Press, 2010.

[3] M. Abramowitz and I. A. Stegun. *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Dover Publications, 1965.

[4] E. A. Coddington and N. Levinson. *Theory of Ordinary Differential Equations*. McGraw–Hill, 1955.

[5] A. Zettl. *Sturm–Liouville Theory*. American Mathematical Society, 2005.

[6] G. B. Arfken, H. J. Weber, and F. E. Harris. *Mathematical Methods for Physicists*. 7th edition, Academic Press, 2013.